

Snowledge

USABILITY TEST REPORT

Snowledge version 2.0.4

11/2023

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1. TEST SUMMARY

This usability test was conducted on week 46 with five participants through the completion of test tasks. Participants were asked to speak aloud, while operating in Snowledge app on Android device, based on test leader's instructions. There were 23 test tasks in total. At the end, link to a survey was sent to the participants, to help understand their overall view of the app. End survey summary is included in Chapter 2 of this document.

Table of test users and test completion

User ID	Test task completion	Failed tasks → (test leader had to help)	Skiing frequency
T1	91 %	1. Going back to Pallas map/starting page (went to map view). 2. Finding info about Pallaksen Pöllöt (skipped first page and searched menu)	Very rarely
T2	100 %	-	Every couple of years
T3	100 %	-	Very rarely
T4	95 %	1. Going back to Pallas map/starting page (went to map view).	Never
T5	100 %	-	Never

Table of identified problems

Task ID	Task	Where	Problem	Criticality (High, Medium, Low)	Frequency (once = 0,2 all = 1,0)	Tester quote or action (translated from Finnish)
1	Rescue	Chat-window	Pressing "Close" when sending chat-message	High	0,4	T2: "I think I just sent it [help-message]. Oh, I think I accidentally pressed "close". Close button was so big...
2	Rescue	Chat-window	Information in chat is always in English e.g., rescuee & helper, close, enter message.	High	-	Nobody mentioned the problem because every test user had good English-skills. This might not be the case with every user in northern Finland.

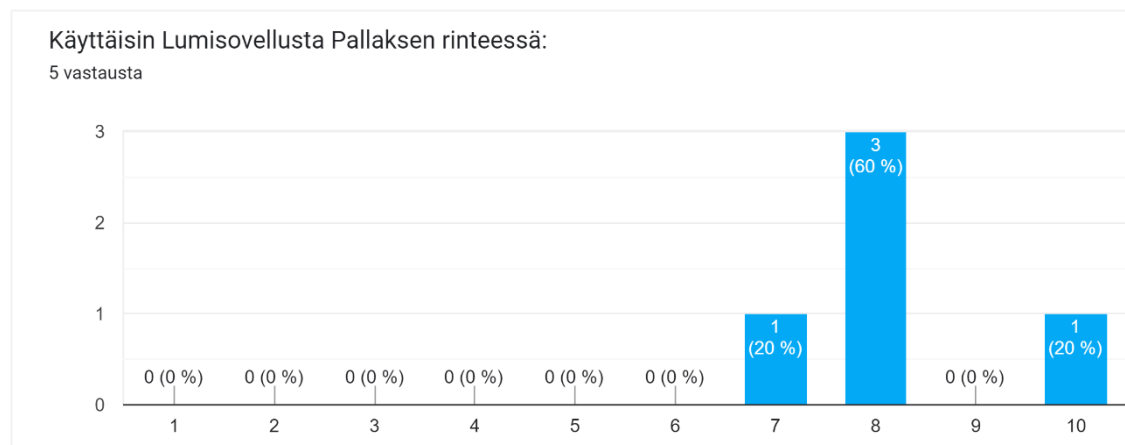
3	Rescue	Rescue map	Text "avunantaja" (in Finnish version) has very poor contrast in help map, hard to see	High	0,4	T4 and T5 had trouble seeing text, even worse outside, while in fell area (sun shines on screen, snowy weather etc.)
4	Rescue	Accept help-request from notification	Accept/decline have poor contrast & colours for the red-green colour-blind	High	0,4	T3: "The contrast could be better here."
5	Snow conditions in Pallas	Menu	Opening "Map view", when asked to go to starting screen (Pallas map)	High	0,6	T1: "I don't know why this is now a map of Tampere. [After search and finding right map] I was confused between snow conditions [in Pallas] and snow type definitions."
6	Rescue	Ask for help-screen	After stopping help request, screen with 112 & further request stays open	Medium	0,2	T1: "This still shows the help request screen... I don't think it is necessary after just stopping to ask for help".
7	Rescue	Rescue-chat	Message icons are always in same alignment, rescuee on the left	Medium	-	Nobody mentioned this, but it would be clearer to have user's own icon on their side, on right. Maybe even on right corner.
8	Rescue	Rescue-chat	Messages are hard to see until phone keyboard is closed, even after pressing send	Medium	0,2	T4: How can I see it [the message]. How can I close the phone keyboard?
9	Snow conditions in Pallas & map view	Pallaksen Pöllöt-logo	Intuitive to click on site/app logo to find information, current logo is inactive	Medium	0,6	T1: "I have tried to click on the logo many times, at least that doesn't work [trying to find info about Pallaksen Pöllöt]".
10	Snow conditions in Pallas	Pallas map	Red background and no text, when erasing highlight of "laskualueet" or snow type	Medium	0,2	T1: "Just my mid-comment, colourblind people might not notice that trick [to erase highlighting only by colour]".
11	Pallaksen Pöllöt-site	Starting screen	Text is very small	Low	0,4	T1 first skipped over opening page's text and started looking for "about us". T5 had to use reading classes. Still, not crucial for the working of Snowledge-app side.
12	Rescue	Accept help-request screen	Before accepting help request, user doesn't know about the problem/might not be able to help	Low	0,2	T2: "But how do I know at the beginning, if it is a problem where I am not able to help?"
13	Snow conditions in Pallas etc.	Pallas map	Many functionalities are only in Finnish (snow conditions, definitions, weather, about the service)	Low	-	Only the Finnish side was used in the usability testing, but to make the app more accessible for e.g., tourists, English translations should be added in the future.

14	Snow conditions in Pallas	Pallas map	It is not clear, what eye-icon does	Tutorial already in the works	0,2	T2: "I pressed the eye-icon, I am not sure what it does."
15	Rescue	Rescue map	Center-button is sometimes hard to notice	Map UI updates in the works	0,2	T3: I think it [centering user] was not clear right away.

2. END SURVEY SUMMARY

Participants filled an end survey after completing the test tasks. End survey questions were agreeable statements and open-ended questions, to better explain given numeric answers. This chapter includes the graphs from Google Form survey. Still, it is good to remember that the test sampling was small (five participants) and personal views affect given answers. From the scale, 1 means negative answer/no, and 10 means positive/yes.

Q1: I would use Snowledge while skiing in Pallas. (1-10)

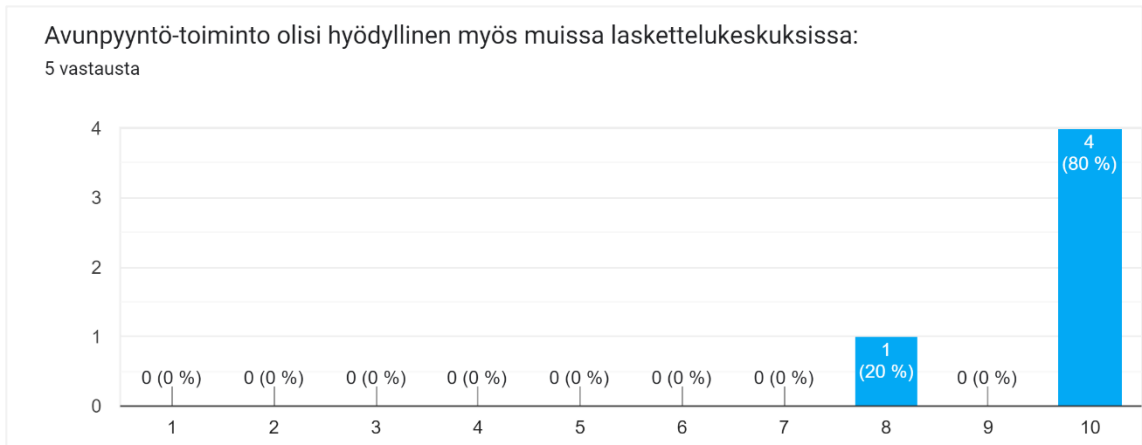


This question was to get participants view on the usefulness of the app, to use it in real-life. Any of the participants weren't active skiers, so the question is more hypothetical. Still, it is possible to see that the answers were mostly positive, one participant would use the app with full certainty (10 out of 10 score).

Some open-ended replies complimented the apps diverse functionalities, weather, snow conditions and rescue-function. It was mentioned that the app lowers the threshold for

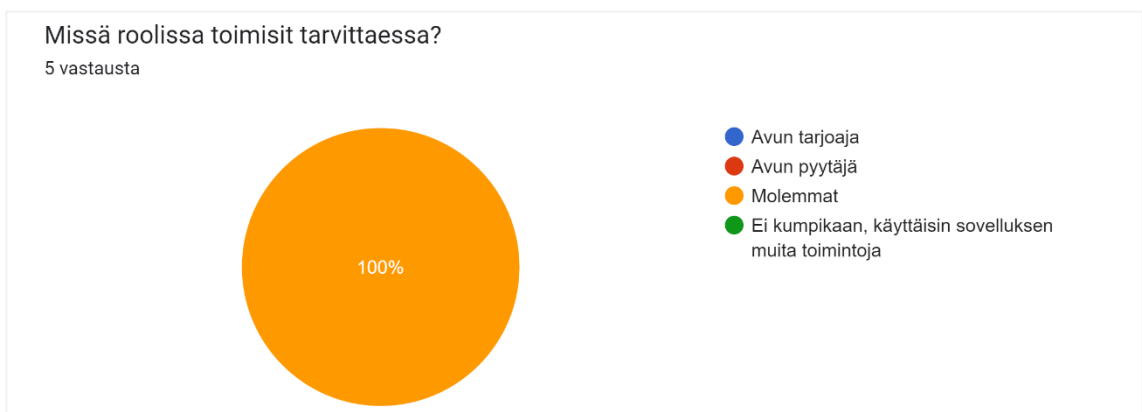
asking help, when 112 is not the only option for receiving it. One participant knew themselves to be inexperienced on fell area/skiing in general, and thus felt like this app could help feeling more secure, if something unexpected were to happen.

Q2: Rescue-function would be useful in other ski-resorts. (1-10)



This question helped to gather information, how the participants see the app and rescuing benefiting other skiers, outside of their own personal use. As the participants weren't active skiers themselves, they agreed on the app's offered value for intended target group. Answers on this statement were very unanimous, 4 out of 5 participants giving it a 10. Rescue-function was seen as very positive addition to Pallas' ski experience, and same kind of communal security net could be seen helpful also elsewhere. This, small sampling result, would also support possible plans of extension beyond Pallas area.

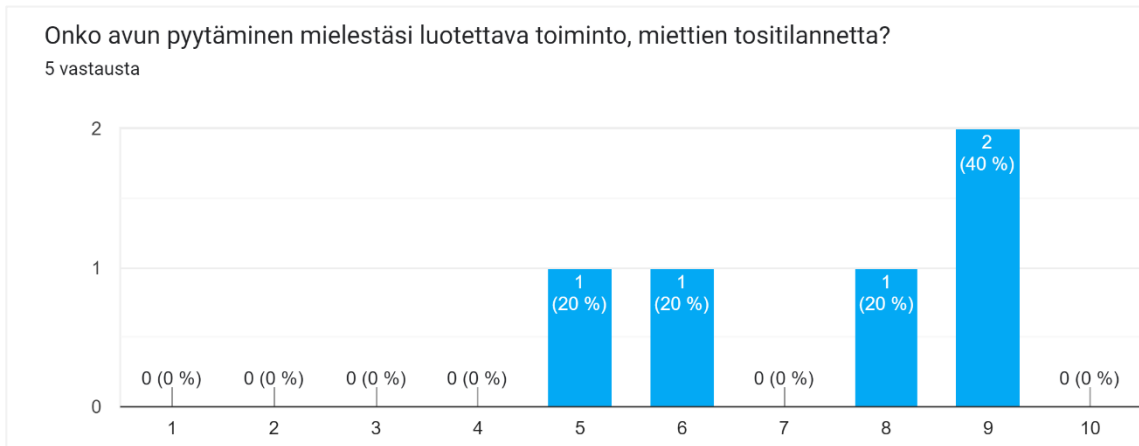
Q3: In what role would you be willing to operate as?



When asking test users, in which role (help requester, -provider, both or neither) they were willing to operate, common answer is clear. All five participants answered "molemmat" meaning both roles. Nobody answered that they wouldn't use rescue-

functionality at all, and would only use e.g., snow condition and weather information. This tells about a sense of care for one another, even strangers.

Q4: Do you think asking for help is reliable, when thinking about real-life situations? (1-10)

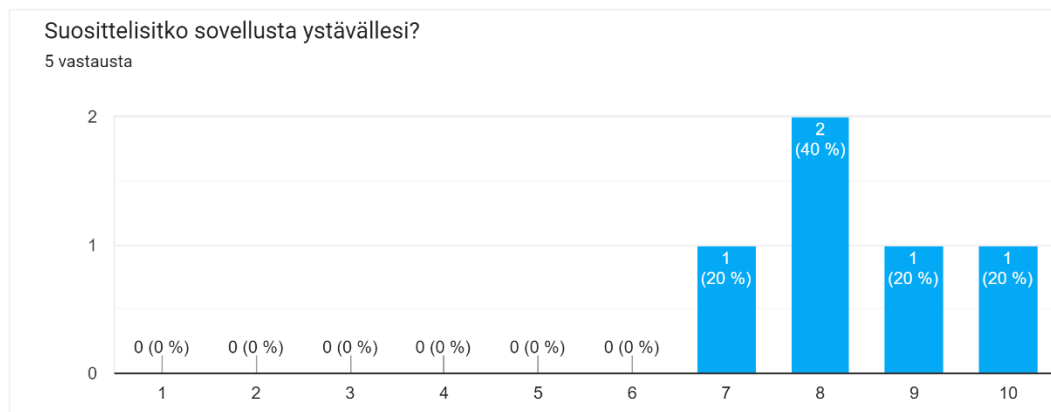


Reliability of the app was the most dividing question on the survey. It was meant to get overall sense of security relating to the rescue-functionality, regarding both humane issues as well as app issues. Common reliability issues mentioned were e.g., loss of internet connection, battery, and cold weather (phone freezes), so normal issues regarding the use of mobile phones in bit more rural area.

Rescue-functionality was seen reliable in its basic overview – every test task relating to the use of it passed efficiently, and feedback on the user flow was overall positive. Help button is easy to notice. Some small feedback was to further hone the map and chat-elements to look more polished. One test user also mentioned even more vibrant use of colours, while still taking account the colourblind (e.g., red-green components must have also always text).

One tester with lower scoring had skepticism about the humane side of rescue-function, especially the standby effect. Sometimes it is easy to assume that somebody else will help, easily leading that nobody will. This can be prevented by seeing the active number of helpers as a helper themselves. Also, seeing the problem type beforehand could prevent dropping the help later, if it is an issue that the person can't assist in.

Q5: Would you recommend Snowledge to your friend? (1-10)



Recommending to friend is a common question in any survey, and thus also here as a last quantitative question. Results were rather positive, indicating the satisfied verdict among the participants.

Last mentions from participants were e.g., reminder to keep the app as simple as possible with given functions, because efficiency and simplicity are advantages in possible bad weather conditions. Also, learning to use all functions could take its time, but tutorial is being implemented to advance this possible issue. To assemble all positive and negative feedback, the app is certainly being implemented in the right direction. Improving already existing functionalities further, while taking account accessibility and different types of users is a key component on honing Snowledge to be the best possible version it can be.